# Encoder-Free Multiaxis Physics-Aware Fusion Network for Remote Sensing Image Dehazing

Yuanbo Wen, Tao Gao, *Member, IEEE*, Jing Zhang, *Member, IEEE*, Ziqi Li, and Ting Chen

***Abstract*— Current methods for remote sensing image dehazing confront noteworthy computational intricacies and yield suboptimal dehazed outputs, thereby circumscribing their pragmatic applicability. To this end, we propose encoder-free multiaxis physics-aware fusion network (EMPF-Net), a novel EMPF-Net that exhibits both light-weighted characteristics and computational efficiency. In our pipeline, we contend that conventional u-shaped networks allocate substantial computational resources to encode haze-degraded features, which play a subordinate role in the reconstruction process. Consequently, our encoder stages solely incorporate down-sampling operations. To improve the representation efficiency and enhance the generalization capabilities, we devise a multiaxis partial queried learning block (MPQLB) that primarily concentrates on learning dimension-wise queries, instead of relying solely on strictly correlated content of the input features. Furthermore, we augment the reconstruction procedure by incorporating ground truth supervision into each stage via a supervised cross-scale transposed attention module (SCTAM). It calculates attention maps under the guidance of clean images, thereby suppressing less informative features to propagate to the subsequent level. In addition, to address the challenge of ineffective intralevel feature fusion, which result in insufficient elimination of haze-degraded information and have a negative impact on the quality of reconstructed images, we introduce a physics-aware intralevel fusion module (PIFM). This module harnesses a physical inversion model to facilitate the intralevel feature interaction and alleviate the interference of dehazing-irrelevant information. Our proposed EMPF-Net is evaluated on 12 publicly available datasets, and the experimental results substantiate our superiority in terms of both metrical scores and visual quality, despite being equipped with a modest parameter count of 300k. Our approach is readily accessible at https://github.com/chdwyb/EMPF-Net.**

***Index Terms*— Computer vision, encoder-free network, image dehazing, physics-aware fusion, remote sensing image.**

## I. Introduction

THE examination and interpretation of remote sensing images occupy a crucial position in diverse domains, including climate prediction, agricultural yield assessment, natural calamity surveillance, mineral resource exploration, and military surveillance [1], [2], [3], [4]. However, remote sensing images acquired under hazy conditions inevitably exhibit diminished contrast and visibility, rendering them unsuitable for subsequent applications in the realm of computer vision. Therefore, the development of an efficient and effective dehazing technique emerges as a pragmatic imperative, promising to enhance the enduring quality of remote sensing images.

Recently, learning-based algorithms for remote sensing image dehazing have gained prominence, capitalizing on advancements in deep learning methodologies, and have demonstrated commendable performance. Among these methods [1], [9], [10], [11], [12], [13], [14], u-shaped architectures are commonly utilized as foundational frameworks to eliminate remote sensing haze. These u-shaped architectures comprise feature encoding and decoding networks, where the decoding phase facilitates the exchange of intralevel information with the encoding phase through the utilization of skip connections. For instance, DMENet [11] introduces a multiscale encoder and a dynamic mutual decoder to improve the robustness and generalization performance. However, existing methodologies tend to adopt similar designs for their encoders and decoders, thereby resulting in substantial computational overhead and overlooking the pivotal role of skip connections in regulating information flow. In addition, with the excellent characterization capability of vision transformer (ViT), most remote sensing fog removal algorithms [1], [9], [15], [16] incorporate ViT as the fundamental building block. For instance, DehazeFormer [9] presents a novel approach by introducing a modified normalization layer and a spatial information aggregation scheme. Notwithstanding the notable advancements made by these methodologies, substantial challenges persist in achieving a delicate equilibrium between the computational efficiency and the quality of the generated image. This is especially evident when addressing high-resolution remote sensing hazy images, where the twin imperatives of effectiveness and efficiency assume paramount importance for practical application.

In this work, we introduce an innovative encoder-free multiaxis physics-aware fusion network (EMPF-Net). As Fig. 1 shown, it achieves the best trade-off between the quality of generated haze-free images and the inference time to process hazy images. Specifically, conventional encoder–decoder networks predominantly employ substantial encoder blocks to

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generate encoding features. We argue that this approach necessitates extensive computations for learning haze-degraded features, rendering the dehazing framework computationally intensive. Therefore, our proposed EMPF-Net deviates from this paradigm by adopting an encoder-free architecture that emphasizes information flow and degradation-free feature reconstruction. In our pipeline, the encoding phase exclusively involves down-sampling operations. To construct basic learning blocks, we draw inspiration from the works of Jose Valanarasu et al. [6], Chen et al. [17], Ruan et al. [18], Carion et al. [19], and Gao et al. [20]. We present a novel multiaxis partial queried learning block (MPQLB), which examines input features along the three dimensions of channel ($c$), height ($h$), and width ($w$). Unlike existing basic blocks, our MPQLB dissects the input features into four components. Subsequently, we apply dimension-wise queries on the $hw$, $ch$, and $cw$ planes to query the three quarter components, respectively. By employing these queries, we regulate the information flow to the multilayer perceptron (MLP) without directly learning the content of the input features, leading to an improvement on generalization capabilities. MPQLB only utilizes $1\times1$ convolution to conduct channel-wise transforming of the left one quarter features, achieving partial self-relevant representation. Verified by the extensive experiments and the works of Jose Valanarasu et al. [6], Carion et al. [19], and Gao et al. [20], our proposed MPQLB not only yields a significant reduction in computational complexity but also enhances the generalization capacity due to its weakly input-related learning strategies.

Moreover, it is noteworthy that the prevailing approaches [1], [9], [15], [21], [22], [23] employed for haze removal in remote sensing images solely involve direct prediction of haze-free images during the final stage. Obviously, these methods overlook the significance of curtailing the influence of haze degradation in the early reconstruction stages. To overcome this limitation, we introduce a fresh attention mechanism that harnesses ground truth supervision to calculate attention maps, suppressing the less informative features and enabling the selective propagation of dehazing-relevant features to the subsequent levels. We propose a supervised cross-scale transposed attention module (SCTAM), which leverages supervised features to derive the cross-scale query representation $\mathbf{Q}_{\text{cross}}$ and key representation $\mathbf{K}_{\text{cross}}$. Furthermore, we incorporate transposed self-attention [24], [25] as it offers the advantage of bringing down the computational complexity and mitigating the issues of feature misalignment caused by scale modifications. Meanwhile, we employ depth-wise convolution to facilitate spatial information extraction and enhance high-frequency modeling capabilities [9], [24], [26]. The value representation $\mathbf{V}$ is acquired through a $1\times1$ convolution and subsequently augmented by the supervised attention maps. As a result, less informative features are suppressed, while the essential features are allowed to propagate to the next feature level.

In addition, to address the limitations of existing intralevel feature fusion strategies, such as $1\times1$ convolution and selective kernel fusion [9], which fail to fully capitalize on the physical properties of haze degradation, leading to suboptimal representations and limited generalization capabilities, we develop a pioneering physics-aware intralevel fusion module (PIFM). In our PIFM, we tailor the network structures to align with the characteristics of the estimated parameters by inversely modeling the haze-degraded images [27]. This approach derives accurate aggregated features that diminish degradation information while preserving essential features. To be specific, we employ the combined features to determine the joint parameters of the physical inversion model, subsequently enhancing and amalgamating the two groups of features originating from the two distinct sources. By virtue of our proposed PIFM, we effectively prevent the transmission of haze-degraded information to the decoding and reconstruction levels, thereby facilitating superior feature fusion and haze elimination of our dehazing framework on remote sensing hazy images.

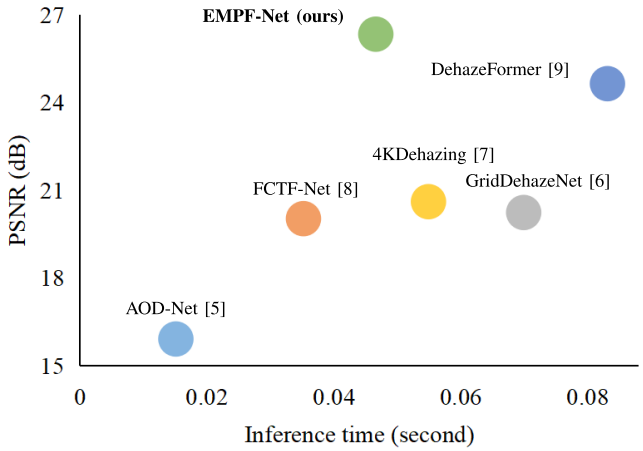



Fig. 1. Comparative analysis of efficiency in remote sensing image dehazing. Our EMPF-Net achieves the best trade-off between the quality of generated haze-free images and the inference time to process hazy images when compared with other image dehazing approaches.

Fig. 2 provides an overall visual representation of our proposed EMPF-Net for dehazing remote sensing images. The primary contributions of our work can be succinctly summarized as follows.

1) We propose an innovative encoder-free design that focuses on information flow and degradation-free feature reconstruction, instead of the calculation-heavy feature encoding networks. This architectural design allows our EMPF-Net to achieve the current best performance on 12 public datasets while requiring only 300k parameters.
2) An intricately designed MPQLB is proposed, integrating dimension-wise queries to conduct partial learning approach, which ameliorates computational overhead while concurrently enhancing the generalization capacity.
3) An SCTAM is introduced, in which ground truth supervision is incorporated during the early reconstruction stages to mitigate the negative impact of less informative features.
4) An innovative PIFM is presented, which governs the information flow based on the inverse haze-degraded

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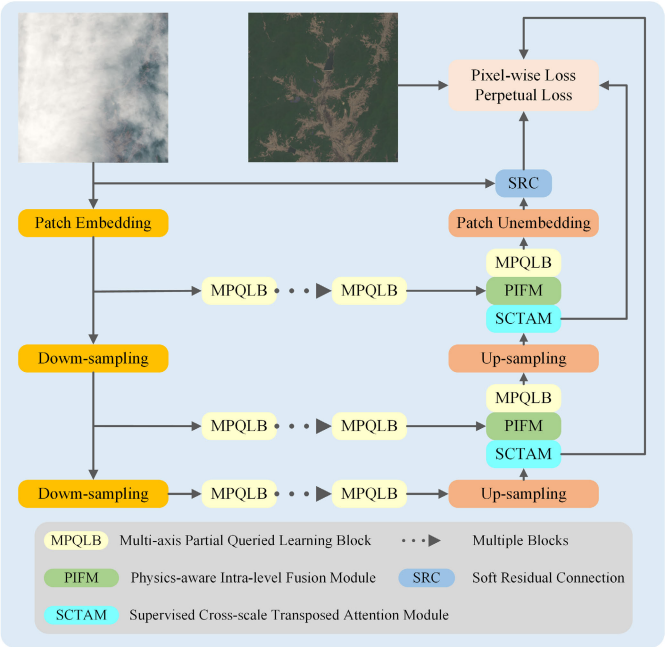



Fig. 2. Architectural overview of our proposed EMPF-Net for remote sensing image dehazing, named EMPF-Net. Our approach centers around a novel u-shaped framework that places emphasis on information flow and feature reconstruction, where the encoder phase is streamlined and consists solely of down-sampling operations. Patch embedding and unembedding are formed by $3 \times 3$ convolution. The key components of EMPF-Net include our proposed MPQLB, SCTAM, and PIFM. By leveraging these novel modules, EMPF-Net achieves a remarkable feat of being a light-weighted network, requiring only 300k parameters.

physical model and facilitates the efficient feature aggregation.

## II. Related Work

### A. Model-Based Remote Sensing Image Dehazing

Model-based approaches utilize DCP [28], color attenuation prior [28], and haze-line prior [29], [30], [31] to address distinct haze-degraded scenarios. However, these methods introduce transmission estimation errors as a result of relying on imprecise prior information. He et al. [28] observe that individual pixels in RGB color channels within nonsky regions exhibit remarkably low values. This discovery enables the transmission estimation to be considered reliable, despite the potential drawback of introducing color distortion in the sky region. Subsequently, Zhu et al. [32] establish a mathematical equation that takes into account the disparities between image brightness and saturation, thereby facilitating the accurate estimation of transmission map. Berman et al. [29] and [30] put forward a nonlocal dehazing method, where the colors of haze-free images form tight clusters within the RGB color space, and haze-degraded images present haze lines. As a result, they estimate the transmission map by analyzing the haze lines. In contrast, an iterative dehazing method designed for single remote sensing images, as outlined by Xu et al. [33], introduces the concept of "virtual depth" to effectively quantify Earth surface coverings. Through an iterative procedure, this approach improves the estimation of transmission and subsequent haze removal. Recently, He et al. [34] introduce a heterogeneous atmospheric light prior and a side window filter for remote sensing image dehazing.

### B. Learning-Based Remote Sensing Image Dehazing

Conventional model-based methods struggle to adapt to the complex composition of haze degradation in remote sensing images, leading to a subpar performance. In contrast, Gu et al. [35] propose a prior-based dense attentive network. This network leverages dense blocks and attention blocks to effectively recover haze-degraded remote sensing images, addressing the challenges posed by the complex haze degradation. Likewise, Trinity Net [1] incorporates prior information into the gradient-guided Swin Transformer [36] for reasonable estimation of haze parameters. Subsequently, FCTF-Net [8] is introduced as a two-stage dehazing network. In the first stage, it focuses on the coarse dehazing to obtain initial results, and in the second stage, it refines the coarse dehazing results to achieve enhanced performance. Likewise, Huang and Chen [37] propose DCRD-Net, a dual-step cascaded residual dense network explicitly tailored for dehazing single remote sensing images. To handle nonuniform haze degradation, Jiang et al. [38] present a deep network that addresses nonuniform haze degradation by combining discrete wavelet transform with a deep learning network while proposing KFA-Net [2], which extends the temporal and spatial scopes of remote sensing image applications, offering a comprehensive solution for effectively mitigating haze with varying degrees of degradation. Additionally, Chen et al. [39] leverage multiple scales to enhance the dehazing performance, accommodating various levels of haze degradation in remote sensing images. Li et al. [21] incorporate a multimodel joint estimation module and a self-correcting module, enabling the construction of a unified end-to-end dehazing framework. The multimodel joint estimation module enhances the estimation accuracy, while the self-correcting module refines the results, leading to improved dehazing outcomes. Recently, DehazeFormer [9] presents a novel approach by introducing a modified normalization layer and a spatial information aggregation scheme. In addition, ARDD-Net [40] develops an adaptive region-based diffusion model to eliminate the remote sensing haze and achieves outstanding performances on both synthetic and real-world hazy images.

## III. Proposed Method

### A. Overall Encoder-Free Pipeline

In the current domain of image dehazing, the prevailing trend is toward employing u-shaped architectures [7], [9], [12], [41], [42], [43], which heavily rely on encoder–decoder networks. However, such architectures impose significant computational demands and resource requirements, especially during the encoding phase, where haze-degraded features are represented through remarkable computational complexity. Furthermore, researchers frequently encounter challenges pertaining to the equitable design of encoders and decoders. In this context, the consideration of whether to employ an equal number of encoders and decoders poses a notable dilemma.

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As illustrated in Fig. 2, we introduce the EMPF-Net, a novel architecture, that adopts a distinctive approach by exclusively incorporating down-sampling operations during the encoding phase, effectively eliminating the heavy need for multiple encoders in the dehazing network. Moreover, our approach incorporates innovative skip connections to effectively regulate the flow of features to the reconstruction stages, surpassing conventional methods such as simple $1 \times 1$ convolutions and selective kernel fusion [9]. Within our pipeline, when presented with a remote sensing hazy image $\mathbf{I} \in \mathbb{R}^{3\times h\times w}$, we leverage the overlapping patch embedding to generate high-dimensional features. We introduce a light-weighted MPQLB to extract the down-sampled features and perform as the basic learning block, employing dimension-wise queries for effective partial feature learning and information flow. In the decoding and reconstruction phase, we incorporate ground truth supervision at each feature level through our SCTAM. It employs supervised cross-scale query representation $\mathbf{Q}_{\text{cross}}$ and key representation $\mathbf{K}_{\text{cross}}$ to calculate the supervised attention maps, which selectively suppress less informative features at the current level, allowing only the useful ones to propagate to the subsequent level. In addition, we propose a PIFM to conduct feature fusion between skip connections and the decoding features. To obtain the consequent dehazed images, we apply a $3 \times 3$ convolution and employ the soft residual connection (SRC) [9]. Experimentation on our EMPF-Net demonstrates outstanding performance with only 300k parameters, striking an optimal trade-off between computational efficiency in haze degradation elimination and the quality of dehazed remote sensing images.

### B. Multiaxis Partial Queried Learning Block

In contrast to most existing remote sensing dehazing methods, which heavily rely on the structures consisting of full-channel convolution [5], [33], [44], [45], [46] as the basic learning block, we have observed that these approaches introduce significant computational complexity and strictly learn the input-relevant information. Consequently, their real-world application is limited. Inspired by the works of Chen et al. [17] and Ruan et al. [18], we have identified that applying similar feature transformations to all feature channels leads to a surplus of redundant information. This redundancy presents two major challenges. Firstly, it hampers the representation ability of dehazing networks to deeply learn nonrepetitive features, which is crucial for improving the generalization ability. Secondly, the excessive redundant information interferes with the subsequent decoding and reconstruction procedures.

To overcome the shortcomings of current algorithms [1], [9], [40] for remote sensing image dehazing, which have a high computational cost and only learn features closely related to the input features, we introduce a light-weighted MPQLB as Fig. 3 shown. In this block, we discern the three quarter of input features along the dimensions of channel ($c$), height ($h$) and width ($w$). For each feature group, we apply learnable queries on the $hw$, $ch$, and $cw$ planes, respectively. A distinctive feature of our approach is the refusal to directly learn the content of input features. Instead, we employ three

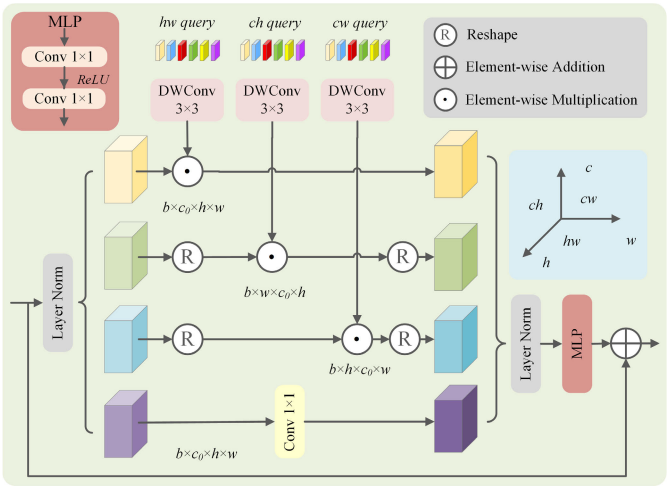



Fig. 3. Illustration of our proposed MPQLB. We utilize axis-oriented queries to regulate the information flow and generate weakly correlated representations with the input features. This technique substantially alleviates the computational burden, making the image dehazing process more efficient, while leading to notable enhancements in performance and generalization ability for remote sensing image dehazing.

groups of queries to conduct partial feature learning and control the information flow into the feed-forward network. Given an input feature matrix $\mathbf{X}$ structured as $(b, c, h, w)$, we partition it into four feature groups, $x_1, x_2, x_3$, and $x_4$. Each group conforms to the structure $(1, c/4, h, w)$. Let $x_0 = c/4$. The configuration of the learnable dimension-wise queries for $x_1$ takes the form of $(b, 1, x, y)$, for $x_2$ it becomes $(b, 1, c_0, x)$, and for $x_3$ it becomes $(b, 1, c_0, y)$, where $x$ and $y$ correspond to the dimensions of queries in the $h$- and $w$-axes. All the queries are randomly initialized separately. When conducting multidimension partial queried learning, we initially transform $x_1, x_2$, and $x_3$ into $(b, c_0, h, w)$, $(b, w, c_0, h)$, and $(b, h, c_0, w)$, respectively. Subsequently, we utilize bilinear interpolation to adjust the last two dimensions of the queries to match the feature groups. Meanwhile, we employ depth-wise convolutions to enhance the representations of these queries. The queried features are acquired through Hadamard product operations between $\mathbf{X}$ and $\mathbf{Q}$. We segregate the features into different dimensions, $\mathbf{X}_{hw} \in \mathbb{R}^{c\times h\times w}$, $\mathbf{X}_{ch} \in \mathbb{R}^{w\times c\times h}$, $\mathbf{X}_{cw} \in \mathbb{R}^{h\times c\times w}$, and $\mathbf{X}_0$ representing the remaining quarter of features. We multiply the corresponding dimension-wise queries $\mathbf{Q}$hw $\in \mathbb{R}^{c\times h\times w}$, $\mathbf{Q}_{ch} \in \mathbb{R}^{w\times c\times h}$, and $\mathbf{Q}_{cw} \in \mathbb{R}^{h\times c\times w}$ with the respective feature channels. For $\mathbf{X}_0$, we only employ a $1 \times 1$ convolution for channel-wise information interaction, achieving partial self-relevant feature representation. Finally, we concatenate the four groups of enhanced features and employ a simple feed-forward network to perform feature transformation. Therefore, the multiaxis partial queried features are derived from

$$\begin{aligned}
[\mathbf{X}_{hw}, \mathbf{X}_{ch}, \mathbf{X}_{cw}, \mathbf{X}_0] &= \text{LN}(\mathbf{X}) \\
[\mathbf{Q}_{hw}, \mathbf{Q}_{ch}, \mathbf{Q}_{cw}] &= \text{DWConv}(\mathbf{Q}) \\
\mathbf{X}'_{\text{queried}} &= \text{Concat}\left(\{\mathbf{Q}_{ij} \odot \mathbf{X}_{ij}\}, \text{Conv}_{1\times1}(\mathbf{X}_0)\right)
\end{aligned} \tag{1}$$

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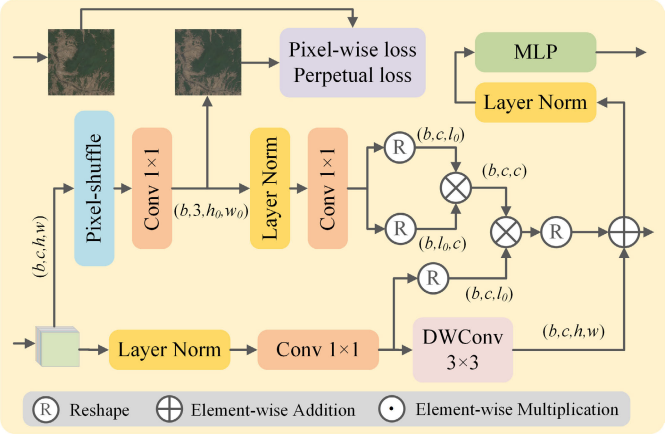



Fig. 4. Illustration of our proposed SCTAM. We provide ground truth supervision to the progressive decoding and reconstruction stages at each feature level, which suppresses the less informative features to propagate to the subsequent level and facilitates our model achieving a significant performance gain.

where LN$(\cdot)$ denotes the layer normalization, DWConv$(\cdot)$ is the depth-wise convolution with kernel size of $3\times3$, Concat$(\cdot)$ indicates the concatenation operation, $\text{Conv}_{1\times1}$ indicates a convolution layer with kernel size of $1\times1$, and $ij \in \{hw, ch, cw\}$.

Furthermore, the output features $\mathbf{X}_{\text{queried}}$ can be formulated as

$$\mathbf{X}_{\text{queried}} = \text{MLP}\left(\mathbf{X}'_{\text{queried}}\right) + \mathbf{X}. \tag{2}$$

The utilization of queries serves a twofold purpose. Firstly, it significantly reduces the computational complexity, making it more efficient and suitable for basic learning block in remote sensing image dehazing. Secondly, the introduction of dimension-wise queries enhances the generalization performance as demonstrated in the works of Jose Valanarasu et al. [6], Ruan et al. [18], and Gao et al. [20]. By refraining from explicitly learning strictly related information of the input feature itself, our MPQLB is better equipped to capture essential patterns and features without getting entangled in redundant or irrelevant information.

### C. Supervised Cross-Scale Transposed Attention Module

Most methods [1], [9], [15], [21], [22], [23] for remote sensing image dehazing predict the dehazed image directly at the final stage, which fail to focus on eliminating haze degradation in the early decoding and reconstruction stages, leading to limited dehazing performance. Instead, we introduce an SCTAM at each feature level, as shown in Fig. 4. The key distinction lies in our multiple considerations of ground truth supervision frameworks. This approach leads to more effective and progressive refinement of haze-free features during the dehazing process, ultimately resulting in higher quality dehazed images. Since features at different levels possess varying resolutions, we employ pixel-unshuffle operations to transform the features into a common resolution space that is consistent with the ground truth. Subsequently, we utilize a $1\times1$ convolution to reduce the number of feature channels to 3 and calculate the losses between the generated cross-scale features and the corresponding ground truth. The generated cross-scale features are formulated as

$$\mathbf{X}_{\text{cross}} = \text{Conv}_{1\times1}(\text{PU}(\mathbf{X})) \tag{3}$$

where PU denotes the pixel-unshuffle operation. Subsequently, the ground truth supervised features are utilized to obtain the attention map following:

$$\begin{aligned}\left[\mathbf{Q}_{\text{cross}}, \mathbf{K}_{\text{cross}}\right] &= \text{Conv}_{1\times1}(\text{LN}(\mathbf{X}_{\text{cross}}))\\ \text{Attn} &= \left(\mathbf{Q}_{\text{cross}} \otimes \mathbf{K}^T_{\text{cross}}\right)\cdot\alpha\end{aligned} \tag{4}$$

where $T$ denotes the matrix transposing operation, Attn indicates the attention map, $\alpha$ is a trainable scaling parameter employed to modulate the dot product between $\mathbf{Q}_{\text{cross}}$ and $\mathbf{K}_{\text{cross}}$. We utilize transposed self-attention mechanism to represent attention maps, which offer a significant reduction in computational burden when compared to spatial self-attention in ViT [9], [47]. Another advantage of transposed self-attention in our SCTAM lies in its spatial-free nature, which allows for the calculation of cross-scale attention maps, facilitating more efficient and effective feature interaction. To further enhance the representation ability of self-attention, we employ depth-wise convolution, aiming at completing the spatial information extraction process and improving the high-frequency modeling ability [9], [24], [26]. Therefore, the features enhanced by our proposed SCTAM are obtained by

$$\begin{aligned}\mathbf{V} &= \text{Conv}_{1\times1}(\text{LN}(\mathbf{X}))\\ \mathbf{X}_{\text{enhanced}} &= \text{FFN}(\text{Attn}\otimes\mathbf{V} + \text{DWConv}(\mathbf{V})).\end{aligned} \tag{5}$$

Our proposed SCTAM offers two significant contributions. Firstly, SCTAM introduces ground truth supervisory at each feature level during the progressive decoding and reconstruction process. These supervisory signals serve as crucial guidance, helping the network refine its representations and aligning them with the ground truth, enabling the model to iteratively improve the quality and accuracy of the generated dehazed images. Secondly, SCTAM leverages locally supervised predictions to generate self-attention maps. These maps are instrumental in identifying and suppressing less informative features at the current level, effectively filtering out irrelevant or redundant information. Therefore, only the most relevant and useful features are allowed, leading to a more focused and effective information flow.

### D. Physics-Aware Intralevel Fusion Module

The atmospheric scattering model [28] is a common formulation used to represent hazy images. It is expressed as $I(x) = J(x)t(x) + A(1 - t(x))$, where $I$ denotes the hazy image, $J(x)$ represents the haze-free image, $t(x)$ is the transmission map, and $A$ indicates the atmospheric light. However, fully model-based approaches often under-perform compared to the learning-based approaches in the context of remote sensing image dehazing, facing challenges in effectively capturing the complex relationships between the haze degradation and background. For intralevel feature fusion in most u-shaped methods [9] for remote sensing image dehazing, conventional methods typically employ $1\times1$ convolution or selective kernel fusion [9]. However, these approaches fail to effectively incorporate

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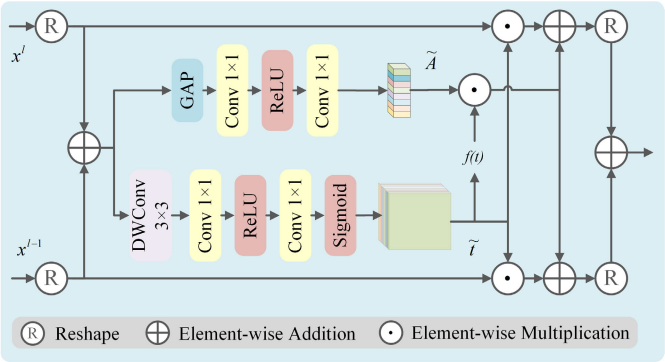



Fig. 5. Illustration of our proposed PIFM. Based on the inverse calculation of the physical haze-degraded model, our proposed PIFM achieves a physics-aware feature fusion that ensures the fusion of intralevel features is consistent with the physical principles governing haze formation and removal.

the physical characteristics of haze degradation during the fusion process. Consequently, the resulting feature representations lack robustness and generalization ability. Inspired by Zheng et al. [27] and Dong and Pan [48], we propose a novel PIFM (seen in Fig. 5, where $f(t) = 1 - t$) that is designed to derive the physical model, effectively integrating the physical characteristics of haze degradation. By incorporating the underlying physics into the fusion procedure, our PIFM facilitates the generation of robust and generalizable representations, resulting in improved performance in remote sensing image dehazing. Initially, we represent the haze-free image $J(x)$ as

$$J(x) = I(x)\frac{1}{t(x)} + A\left(1 - \frac{1}{t(x)}\right) \tag{6}$$

which can be rewritten into

$$\mathbf{J} = \mathbf{I} \odot \frac{1}{\mathbf{T}} + \mathbf{A} \odot \left(\mathbf{1} - \frac{1}{\mathbf{T}}\right) \tag{7}$$

where $\mathbf{J}$, $\mathbf{I}$, $\mathbf{T}$, and $\mathbf{A}$ denote the matrix-vector forms of $J(x)$, $I(x)$, $t(x)$, and $A$, respectively, $\mathbf{1}$ is the identity matrix, $\odot$ indicates the Hadamard product. Subsequently, we utilize kernel $\mathbf{K}$ to extract the features following:

$$\mathbf{K} \circledast \mathbf{J} = \mathbf{K} \circledast \mathbf{I} \odot \mathbf{D} + \mathbf{K} \circledast \mathbf{A} \odot (\mathbf{1} - \mathbf{D}) \tag{8}$$

where $\mathbf{D} = \mathbf{1}/\mathbf{T}$, $\circledast$ denotes the convolution operator. According to the properties of Hadamard product, we have

$$\mathbf{K} \circledast \mathbf{J} = (\mathbf{K} \circledast \mathbf{I}) \odot (\mathbf{K} \circledast \mathbf{D}) + (\mathbf{K} \circledast \mathbf{A}) \odot (\mathbf{K} \circledast (\mathbf{1} - \mathbf{D})). \tag{9}$$

To facilitate model-driven learning-based processes, we introduce $\tilde{\mathbf{A}}$ as a proximate representation of the features $\mathbf{K} \circledast \mathbf{A}$ pertaining to the atmospheric light. Likewise, we define $\tilde{\mathbf{t}}$ as an approximation of $\mathbf{K} \circledast \mathbf{D}$, which captures the transmission map. Moreover, $\mathbf{K} \circledast \mathbf{I}$ and $\mathbf{K} \circledast \mathbf{J}$ correspond to the extracted features of the hazy image and its corresponding haze-free counterpart, respectively.

Regarding the single-source input features $\mathbf{X}$, we learn its associated physics-aware features $\tilde{\mathbf{J}}$ [27] as

$$\tilde{\mathbf{J}} = \mathbf{X}^l \odot \tilde{\mathbf{t}} + \tilde{\mathbf{A}} \odot \left(\mathbf{1} - \tilde{\mathbf{t}}\right) \tag{10}$$

where $l$ denotes the level of input features. In the pursuit of physics-aware fusion, we initially derive the fused features [9], [15] of $\mathbf{X}^l$ and $\mathbf{X}^{l-1}$ as

$$\mathbf{X} = \mathbf{X}^l + \mathbf{X}^{l-1}. \tag{11}$$

Based on (10), we generate $\tilde{\mathbf{A}}$ as follows:

$$\tilde{\mathbf{A}} = \sigma_1(\mathrm{Conv}_{1\times1}(\mathrm{ReLU}(\mathrm{Conv}_{1\times1}(\mathrm{GAP}(\mathbf{X}))))) \tag{12}$$

where $\sigma_1(\cdot)$ is the softmax function following [9], $\mathrm{Conv}_{1\times1}$ denotes the convolution layer with kernel size of $1\times1$, $\mathrm{GAP}(\cdot)$ indicates the global average pooling to eliminate unnecessary information in the homogeneous feature space. To avoid spatial information loss, we utilize depth-wise convolution to enhance spatial features, which is different from [27]. Subsequently, we apply similar operations as described in (12) to derive the nonhomogeneous features $\tilde{\mathbf{t}}$ associated with the transmission map, which can be expressed as

$$\tilde{\mathbf{t}} = \sigma_2(\mathrm{Conv}_{1\times1}(\mathrm{ReLU}(\mathrm{Conv}_{1\times1}(\mathrm{DWConv}(\mathbf{X}))))) \tag{13}$$

where $\sigma_2(\cdot)$ is the sigmoid function, DWConv denotes the depth-wise convolution layer with kernel size of $3\times3$. Consequently, the fused physics-aware features of $\mathbf{X}^l$ and $\mathbf{X}^{l-1}$ are formulated as

$$\begin{aligned}
\left[\tilde{\mathbf{t}}_l, \tilde{\mathbf{t}}_{l-1}\right] &= \tilde{\mathbf{t}} \\
\left[\tilde{\mathbf{A}}_l, \tilde{\mathbf{A}}_{l-1}\right] &= \tilde{\mathbf{A}} \\
\mathbf{X}^l_{\mathrm{fused}} &= \mathbf{X}^l \odot \tilde{\mathbf{t}}_l + \tilde{\mathbf{A}}_l \odot \left(\mathbf{1} - \tilde{\mathbf{t}}_l\right) \\
\mathbf{X}^{l-1}_{\mathrm{fused}} &= \mathbf{X}^{l-1} \odot \tilde{\mathbf{t}}_{l-1} + \tilde{\mathbf{A}}_{l-1} \odot \left(\mathbf{1} - \tilde{\mathbf{t}}_{l-1}\right) \\
\mathbf{X}_{\mathrm{fused}} &= \mathbf{X}^l_{\mathrm{fused}} + \mathbf{X}^{l-1}_{\mathrm{fused}}
\end{aligned} \tag{14}$$

where $\mathbf{X}_{\mathrm{fused}}$ denotes the physics-aware fused features. In contrast to existing approaches, our proposed PIFM utilizes a physics-aware module to generate fused features of the input features from two intralevel sources, avoiding the reception of degenerate information from skip connections, thereby ensuring smooth decoding and reconstruction processes.

### E. Loss Function

Based on our experimental findings, we observe that training our proposed EMPF-Net solely using the commonly used $\mathcal{L}_2$ loss function results in dehazed images with slightly lower structural similarity (SSIM), despite exhibiting superior performance in peak signal-to-noise ratio (PSNR). To address this, we incorporate perceptual loss to measure the dissimilarity between the prediction $\tilde{\mathbf{J}}$ and the ground truth $\mathbf{J}$ in terms of their extracted features. These features are obtained from the 3rd, 8th, and 15th layers of a pretrained VGG16 network, and the perceptual loss is formulated as

$$\mathcal{L}_{\mathrm{perceptual}}\left(\tilde{\mathbf{J}}, \mathbf{J}\right) = \mathcal{L}_2\left(\mathrm{VGG}_{3,8,15}\left(\tilde{\mathbf{J}}\right), \mathrm{VGG}_{3,8,15}(\mathbf{J})\right). \tag{15}$$

Therefore, the whole loss function is

$$\mathcal{L}\left(\tilde{\mathbf{J}}, \mathbf{J}\right) = \mathcal{L}_2\left(\tilde{\mathbf{J}}, \mathbf{J}\right) + \lambda\mathcal{L}_{\mathrm{perceptual}}\left(\tilde{\mathbf{J}}, \mathbf{J}\right) \tag{16}$$

where $\lambda = 0.04$ is utilized to control the contribution of both loss functions following [6], [20].

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## IV. Experimental Results

### A. Experimental Settings

*1) Implementation Specifications:* Our proposed EMPF-Net undergoes a training process of 200 epochs, utilizing a single NVIDIA GeForce RTX 3090 GPU. The patch size and batch size are set to $512 \times 512$ and 10, respectively. For optimization, we employ the Adam optimizer [49] with $\beta_1 = 0.9$ and $\beta_2 = 0.999$, along with an initial learning rate of 0.001. This learning rate is gradually decreased to $1 \times 10^{-8}$ using a cosine annealing scheduler [50]. The number of basic learning blocks is set to (4, 4, 4, 2, 2) for the three skip connections and two decoding blocks, respectively. The numbers of feature channels in the three levels are set to (24, 48, 96). During training process, data augmentation techniques such as random rotation and horizontal flipping are applied. In inference phase, the resolution of input hazy images is adjusted to a square multiple of $128 \times 128$. After that, padding is removed from the recovered images, which ensures that our model can be adapted to various resolutions.

*2) Datasets:* We conduct evaluations of the proposed EMPF-Net using 12 publicly available datasets, each serving distinct purposes and characteristics. StateHaze1K dataset [51] is composed of three subdatasets, namely StateHaze1K-thin, StateHaze1K-moderate, and StateHaze1K-thick. Each subdataset contains 400 pairs of synthetic remote sensing hazy images, distributed as 320 image pairs are used for training, 35 image pairs are employed for validating, and 45 image pairs are reserved for evaluating the performance. RS-Haze dataset [9] is a large-scale dataset comprising 51 300 paired images designated for training and 2700 paired images for testing. Like StateHaze1K dataset, RS-Haze dataset is categorized into three distinct haze levels as thin, moderate, and thick. RICE1 and RICE2 datasets [52] are both collected from Google Earth and designed specifically for remote sensing image declouding tasks. Each dataset comprises a total of 500 paired images. Out of these, 400 image pairs are randomly selected and used for training purposes, while the remaining images are reserved for evaluation and performance assessment. RSID dataset [1] comprises a total of 1000 paired images, where each image has a corresponding hazy and haze-free version. For the purpose of model training, we randomly selected 900 image pairs from this dataset. The remaining 100 image pairs serve as a separate and independent dataset to evaluate how well the model generalizes to unseen data. LHID dataset [53] is designed for dehazing tasks with synthetic light haze. It includes a training dataset and two separate testing datasets, named LHID-A and LHID-B. We merge both datasets, resulting in a combined testing dataset with a total of 1000 image pairs. DHID dataset [53] is tailored for dehazing tasks under dense haze conditions. This dataset contains a total of 14 490 image pairs that are used for model training and includes a separate testing dataset containing 500 image pairs. Dense-Haze dataset [54] serves as the NTIRE 2019 challenge dataset for dense and homogeneous haze. It consists of a total of 45 training image pairs, 5 validation image pairs, and 5 testing image pairs. NH-Haze dataset [55] is designated as the NTIRE 2020 challenge dataset, specifically focusing on nonhomogeneous haze. This dataset consists of a total of 45 training image pairs, 5 validation image pairs, and 5 testing image pairs. As part of the evaluation process, our model is also evaluated on our collected real-world remote sensing hazy dataset, named RRSD300. This dataset comprises a total of 300 hazy images collected from the Microsoft Bing and DIOR dataset [56].

*3) Evaluation Protocols:* In our investigation of supervised hazy datasets, we have adopted the following evaluation methodologies: PSNR, SSIM, and mean squared error (mse). Elevated PSNR and SSIM scores are indicative of improved image quality, while reduced mse values are indicative of enhanced performance of the evaluated techniques. To gauge the effectiveness of the approaches in real-world unsupervised remote sensing image dehazing, we have employed the subsequent metrics: naturalness image quality evaluator (NIQE) [57] and integrated local NIQE (IL-NIQE) [58]. Diminished values of these metrics are associated with superior perceptual image quality.

### B. Synthetic Remote Sensing Image Dehazing

To assess the efficacy of our proposed EMPF-Net in synthetic remote sensing image dehazing, we conduct a comparative analysis with other dehazing methods on multiple datasets, namely StateHaze1K [51], RS-Haze [9], LHID [53], DHID [53], RICE1 [52], RICE2 [52], and RSID [1] datasets. Quantitative evaluation results on StateHaze1K and RS-Haze datasets are summarized in Table I. Our EMPF-Net demonstrates superior performance in terms of PSNR, SSIM, and mse. Specifically, our EMPF-Net outperforms DehazeFormer by 1.49 and 0.48 dB in average PSNR on the two datasets, respectively. For more comprehensive comparisons, we present the visual results and corresponding error maps in Fig. 6. The evaluated methods include DCP [28], AOD-Net [5], GridDehazeNet [59], 4KDehazing [7], DehazeFormer [9], and our proposed EMPF-Net. As illustrated, our EMPF-Net generates images that closely resemble the ground truth, as evidenced by the corresponding error maps (lower brightness corresponds to closer proximity to the ground truth). Remarkably, despite achieving the second-best metrical scores, DehazeFormer consistently produces dehazed images that deviate the farthest from the ground truth counterparts across all evaluated datasets.

As indicated in Table II, our proposed EMPF-Net demonstrates notable performance improvements of 0.93, 1.14, 4.08, 1.45, and 2.34 dB in PSNR on LHID [53], DHID [53], RICE1 [52], RICE2 [52], and RSID [1] datasets, respectively. These findings underscore the superior generalization capabilities of our method, enabling it to achieve excellent performance across diverse datasets. In contrast, other methods struggle to maintain consistent competitiveness across different datasets. Visual comparisons presented in Fig. 7 corroborate our qualitative results, as our proposed EMPF-Net exhibits the highest visual quality and the lowest errors. For instance, our EMPF-Net excels in restoring roads and buildings in the second image, while the results obtained by other methods display noticeable damages. This superior performance is evident both in the visual results and the corresponding error maps.

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TABLE I

REMOTE SENSING IMAGE DEHAZING RESULTS ON STATEHAZE1K [51] AND RS-HAZE [9] DATASETS. IN THIS TABLE, RED AND BLUE INDICATE THE BEST AND SECOND-BEST QUANTITATIVE PERFORMANCE, RESPECTIVELY. OUR PROPOSED EMPF-NET OBTAINS THE BEST PERFORMANCE ON ALL THESE TESTING REMOTE SENSING HAZY DATASETS

| Method | StateHaze1K-thin | | | StateHaze1K-moderate | | | StateHaze1K-thick | | | StateHaze1K-average | | | RS-Haze | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PSNR | SSIM | MSE | PSNR | SSIM | MSE | PSNR | SSIM | MSE | PSNR | SSIM | MSE | PSNR | SSIM | MSE |
| DCP [28] | 13.15 | 0.7246 | 0.0473 | 9.780 | 0.5735 | 0.1072 | 10.25 | 0.5850 | 0.0942 | 11.06 | 0.6477 | 0.0832 | 17.86 | 0.7340 | 0.0201 |
| AOD-Net [5] | 19.54 | 0.8543 | 0.0113 | 20.10 | 0.8854 | 0.0101 | 15.92 | 0.7313 | 0.0265 | 18.52 | 0.8234 | 0.0159 | 27.09 | 0.8476 | 0.0032 |
| GridDehazeNet [59] | 24.34 | 0.9155 | 0.0037 | 23.05 | 0.9274 | 0.0051 | 20.24 | 0.8312 | 0.0096 | 22.54 | 0.8916 | 0.0061 | 35.23 | 0.9446 | 0.0004 |
| FCFT-Net [8] | 23.59 | 0.9127 | 0.0044 | 22.88 | 0.9272 | 0.0055 | 20.03 | 0.8156 | 0.0101 | 22.17 | 0.8852 | 0.0067 | 33.28 | 0.9417 | 0.0007 |
| H2RL-Net [39] | 20.91 | 0.8797 | 0.0083 | 22.34 | 0.9061 | 0.0065 | 17.41 | 0.7684 | 0.0228 | 20.22 | 0.8514 | 0.0125 | 31.18 | 0.9212 | 0.0019 |
| 4KDehazing [7] | 22.35 | 0.9550 | 0.0059 | 23.73 | 0.9565 | 0.0047 | 20.61 | 0.9264 | 0.0088 | 22.23 | 0.9460 | 0.0065 | 33.80 | 0.9442 | 0.0007 |
| M2SCN [21] | 25.21 | 0.9175 | 0.0031 | 26.11 | 0.9416 | 0.0038 | 21.33 | 0.8289 | 0.0095 | 24.22 | 0.8960 | 0.0055 | 37.75 | 0.9497 | 0.0003 |
| DehazeFormer [9] | 26.28 | 0.9477 | 0.0024 | 29.78 | 0.9677 | 0.0012 | 24.64 | 0.9241 | 0.0034 | 26.90 | 0.9465 | 0.0023 | 39.57 | 0.9701 | 0.0002 |
| **EMPF-Net** | 27.40 | 0.9602 | 0.0018 | 31.45 | 0.9748 | 0.0007 | 26.33 | 0.9276 | 0.0023 | 28.39 | 0.9542 | 0.0016 | 40.05 | 0.9767 | 0.0001 |

TABLE II

REMOTE SENSING IMAGE DEHAZING RESULTS ON LHID [53], DHID [53], RICE1 [52], RICE2 [52], AND RSID [1] DATASETS. OUR PROPOSED EMPF-NET OBTAINS PERFORMANCE GAINS OF UP TO 4.08 AND 2.34 DB IN TERMS OF PSNR ON RICE1 AND RSID DATASETS, RESPECTIVELY. MEANWHILE, OUR MODEL ALSO ACHIEVES THE BEST PERFORMANCE ON ALL THESE DATASETS

| Method | LHID | | | DHID | | | RICE1 | | | RICE2 | | | RSID | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PSNR | SSIM | MSE | PSNR | SSIM | MSE | PSNR | SSIM | MSE | PSNR | SSIM | MSE | PSNR | SSIM | MSE |
| DCP [28] | 19.17 | 0.7561 | 0.0213 | 15.86 | 0.7168 | 0.0312 | 16.03 | 0.6966 | 0.0392 | 10.37 | 0.4491 | 0.1031 | 13.33 | 0.7021 | 0.0659 |
| DehazeNet [44] | 21.65 | 0.8821 | 0.0101 | 15.91 | 0.7041 | 0.0302 | 23.74 | 0.8645 | 0.0084 | 20.81 | 0.7742 | 0.0218 | 18.44 | 0.8805 | 0.0202 |
| AOD-Net [5] | 21.15 | 0.8868 | 0.0104 | 15.04 | 0.6674 | 0.0414 | 19.84 | 0.7396 | 0.0127 | 21.16 | 0.7945 | 0.0106 | 18.18 | 0.8777 | 0.0194 |
| GridDehazeNet [59] | 24.98 | 0.9268 | 0.0051 | 25.01 | 0.9090 | 0.0038 | 30.01 | 0.9662 | 0.0015 | 26.20 | 0.8612 | 0.0056 | 23.68 | 0.9326 | 0.0054 |
| FFA-Net [60] | 25.34 | 0.8972 | 0.0042 | 24.03 | 0.8603 | 0.0044 | 31.78 | 0.9543 | 0.0013 | 24.65 | 0/7932 | 0.0063 | 24.05 | 0.8993 | 0.0049 |
| FCTF-Net [8] | 23.85 | 0.9008 | 0.0083 | 18.77 | 0.7935 | 0.0218 | 28.49 | 0.9170 | 0.0046 | 22.58 | 0.8224 | 0.1000 | 19.31 | 0.8559 | 0.0168 |
| 4KDehazing [7] | 31.42 | 0.9522 | 0.0020 | 27.42 | 0.8984 | 0.0054 | 26.43 | 0.9460 | 0.0061 | 26.18 | 0.8569 | 0.0055 | 25.02 | 0.8679 | 0.0040 |
| DehazeFormer [9] | 35.67 | 0.9821 | 0.0004 | 30.19 | 0.9542 | 0.0027 | 31.88 | 0.9622 | 0.0014 | 34.28 | 0.9572 | 0.0009 | 23.77 | 0.8955 | 0.0063 |
| **EMPF-Net** | 36.60 | 0.9837 | 0.0003 | 31.33 | 0.9609 | 0.0011 | 35.96 | 0.9872 | 0.0008 | 35.73 | 0.9704 | 0.0005 | 26.11 | 0.9629 | 0.0031 |

Therefore, the overall superiority of our EMPF-Net, both in quantitative metrics and qualitative visual results, indicates its exceptional generalization ability, making it a promising and reliable solution for synthetic remote sensing image dehazing tasks.

### C. Real-World Natural Image Dehazing

To evaluate the performance of our proposed EMPF-Net on real-world remote sensing image dehazing, we conduct experiments on Dense-Haze [54] and NH-Haze [55] datasets. The experimental results, presented in Table III, demonstrate that our EMPF-Net outperforms all other evaluated methods in terms of quantitative performance on both datasets with complex natural haze degradation. Specifically, EMPF-Net achieves noteworthy improvements of 5.83 and 1.20 dB in PSNR over DehazeFormer on the Dense-Haze and NH-Haze datasets, respectively. Furthermore, our model exhibits the highest performance in terms of SSIM scores. The visual evaluations of our proposed method and the other evaluated methods, depicted in Fig. 8, corroborate the superior performance of our model. Although complete removal of complex haze remains challenging for all assessed methods, our EMPF-Net stands out by providing better visual quality and lower errors. For instance, our approach generates remarkable haze removal results in the second, third, and fourth hazy images, effectively eliminating the haze presence, which is in stark contrast to the noticeable residual haze found in the results of other competing methods.

TABLE III

NATURAL IMAGE DEHAZING RESULTS ON DENSE-HAZE [54] AND NH-HAZE [55] DATASETS. OUR PROPOSED EMPF-NET ADVANCES THE OTHER EVALUATED METHODS IN BOTH PSNR AND SSIM

| Method | Dense-Haze | | | NH-Haze | | |
|---|---|---|---|---|---|---|
| | PSNR | SSIM | MSE | PSNR | SSIM | MSE |
| DCP [28] | 10.06 | 0.3856 | 0.1016 | 10.57 | 0.5196 | 0.0749 |
| DehazeNet [44] | 13.84 | 0.4252 | 0.0743 | 16.62 | 0.5238 | 0.0711 |
| AOD-Net [5] | 13.14 | 0.4144 | 0.0625 | 15.40 | 0.5693 | 0.0589 |
| GridDehazeNet [59] | 13.31 | 0.3681 | 0.0627 | 13.80 | 0.5370 | 0.0390 |
| FFA-Net [60] | 14.39 | 0.4524 | 0.0444 | 19.87 | 0.6915 | 0.0129 |
| MSBDN [61] | 15.37 | 0.4858 | 0.0484 | 19.23 | 0.7056 | 0.0268 |
| KDDN [62] | 14.28 | 0.4074 | 0.0431 | 17.39 | 0.5897 | 0.0300 |
| 4KDehazing [7] | 14.69 | 0.5104 | 0.0420 | 15.39 | 0.6686 | 0.0207 |
| AECR-Net [43] | 15.80 | 0.4660 | 0.0388 | 19.88 | 0.7173 | 0.0135 |
| DehazeFormer [9] | 16.29 | 0.5100 | 0.0079 | 20.47 | 0.7310 | 0.0114 |
| **EMPF-Net** | 22.12 | 0.7391 | 0.0063 | 21.67 | 0.8056 | 0.0083 |

### D. Real-World Remote Sensing Image Dehazing

To evaluate our proposed EMPF-Net on real-world remote sensing haze removal, we conduct experimental comparisons on RRSD300 dataset. To ensure fairness in comparisons, all the reported results on real-world hazy images are obtained by evaluating the corresponding models trained on the StateHaze1K-thick dataset. Table IV presents the qualitative results of the evaluated methods, demonstrating that EMPF-Net achieves the best naturalness scores in NIQE and IL-NIQE, indicating superior quality in haze removal for real-world remote sensing images. Visual comparisons in Fig. 9 further support the effectiveness of EMPF-Net in haze

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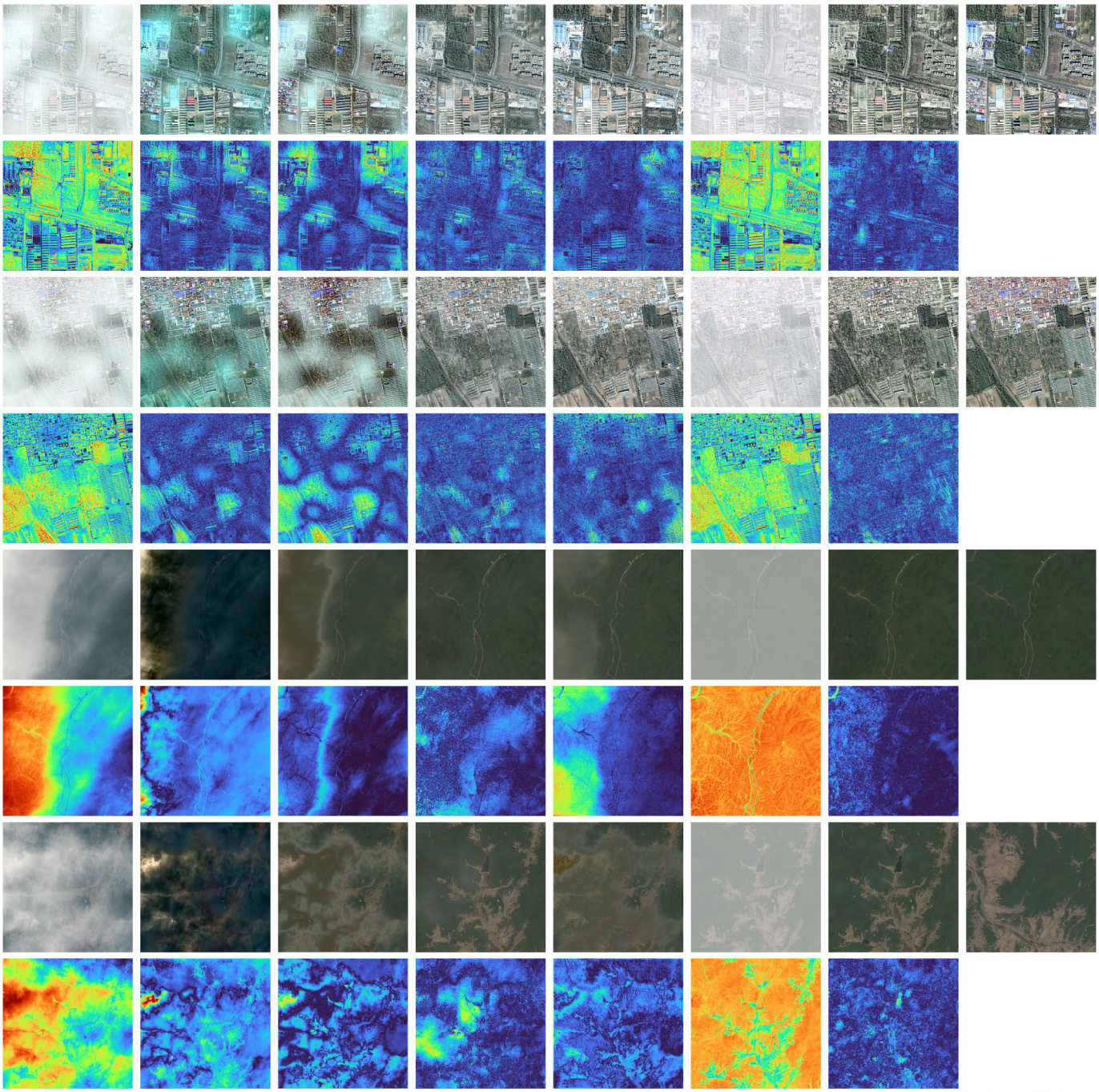



Fig. 6. Visual samples generated by the evaluated methods on StateHaze1K [51] and RS-Haze [9] datasets. In this figure, please note that beneath each hazy or restored image, we present the corresponding error maps depicting the discrepancies between the processed image and the corresponding ground truth. Our EMPF-Net generates images that closely resemble the ground truth, as evidenced by the error maps.

removal for remote sensing images. Our model successfully eliminates haze and better restores image details, outperforming other evaluated methods in this regard. It is noteworthy that, despite DehazeFormer [9] showcasing the second-best performance on most synthetic datasets, it exhibits the weakest performance when applied to real-world remote sensing hazy images.

### *E. Ablation Studies*

The effectiveness of the proposed modules is assessed through ablation studies conducted on the StateHaze1K-thick dataset. These studies serve to analyze the individual contributions of each component within our proposed EMPF-Net and enable comparisons with their similar designs.

*1) Individual Components:* In this study, we employ a baseline architecture with an u-shaped design, utilizing ResNet-Block (RNB) [63] as the basic learning block. Table V depicts the ablation studies on individual components, where the two boxes indicate separate comparisons of the metrics in each. Initially, we make modifications to the pipeline by migrating all encoding blocks to the skip connections, creating an encoder-free framework (EF). This modification keeps

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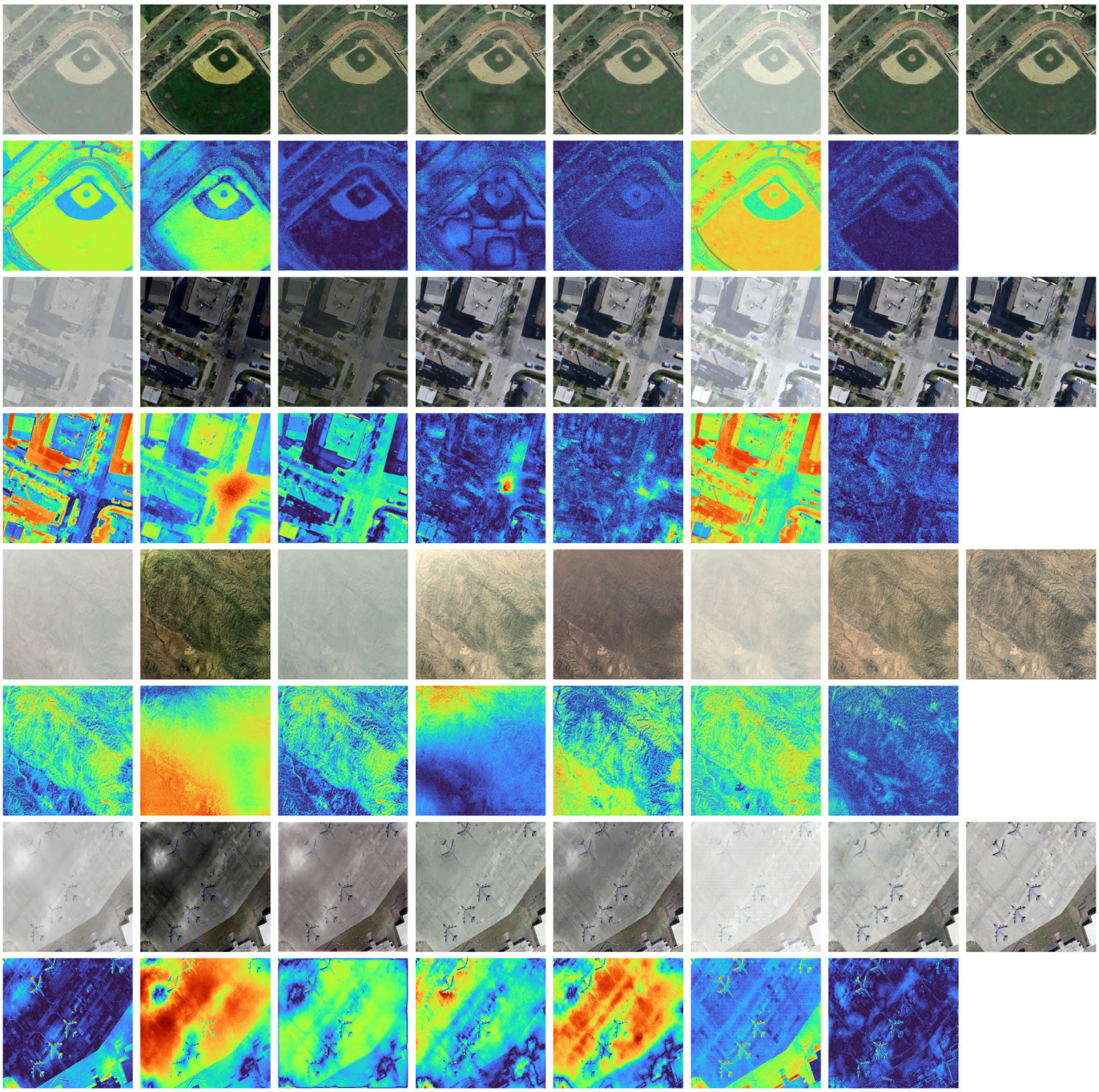



Fig. 7. Visual samples generated by the evaluated methods on LHID [53], DHID [53], RICE1 [52], RICE2 [52], and RSID [1] datasets. Our proposed EMPF-Net exhibits the highest visual quality and the lowest errors.

the number of parameters (#Param) and the floating-point operations (FLOPs) unchanged. Experimentally, EF achieves a performance gain of 0.31 dB in PSNR over the commonly used encoder–decoder architecture, while maintaining the same computational burden. Subsequently, we replace the RNB with our proposed MPQLB, resulting in a substantial improvement of 1.45 dB in PSNR. Remarkably, this improvement is achieved with only 21.59% #Param and 21.49% FLOPs of the baseline model, demonstrating the efficiency and effectiveness of the MPQLB. Meanwhile, MPLB (nonqueried MPQLB) obtains even worse performance than baseline, which indicates that our dimension-wise queries are effective. After incorporating our proposed SCTAM, the model achieves an additional performance gain of 0.93 dB, albeit with a slight increase in computational burden. This result indicates the capability of our SCTAM in improving performance and demonstrates the significance of cross-scale ground truth supervision. Finally, we replace the intralevel feature fusion module with our PIFM, which leads to an improvement of 0.44 dB in PSNR while slightly reducing #Param and FLOPs. In conclusion, the ablation study on individual components demonstrates the significant impact of each proposed individual component in a

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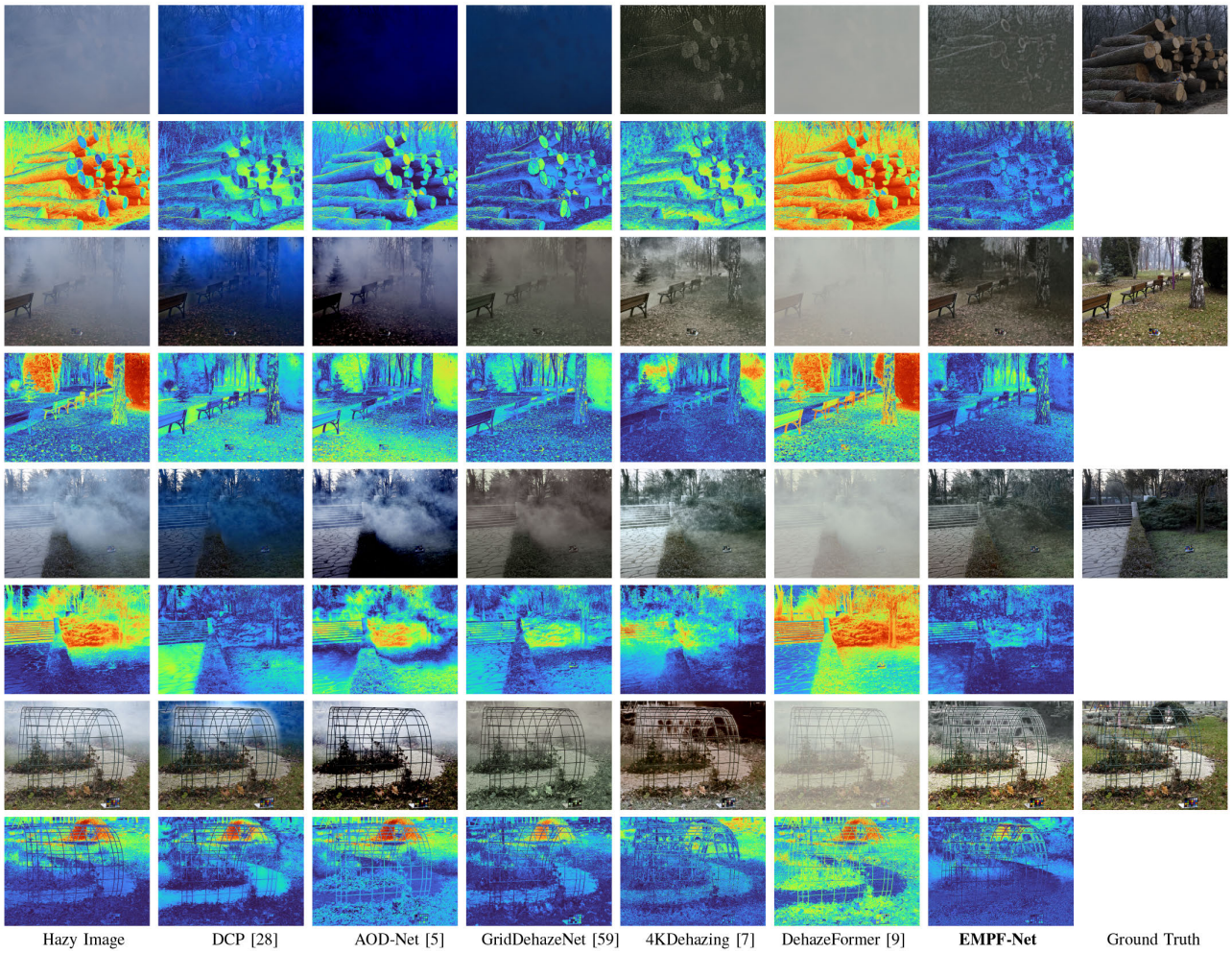



Fig. 8. Visual samples generated by the evaluated methods on Dense-Haze [53] and NH-Haze [55] datasets. Although complete removal of this haze remains challenging for all assessed methods, our EMPF-Net stands out by providing better visual quality and lower errors.

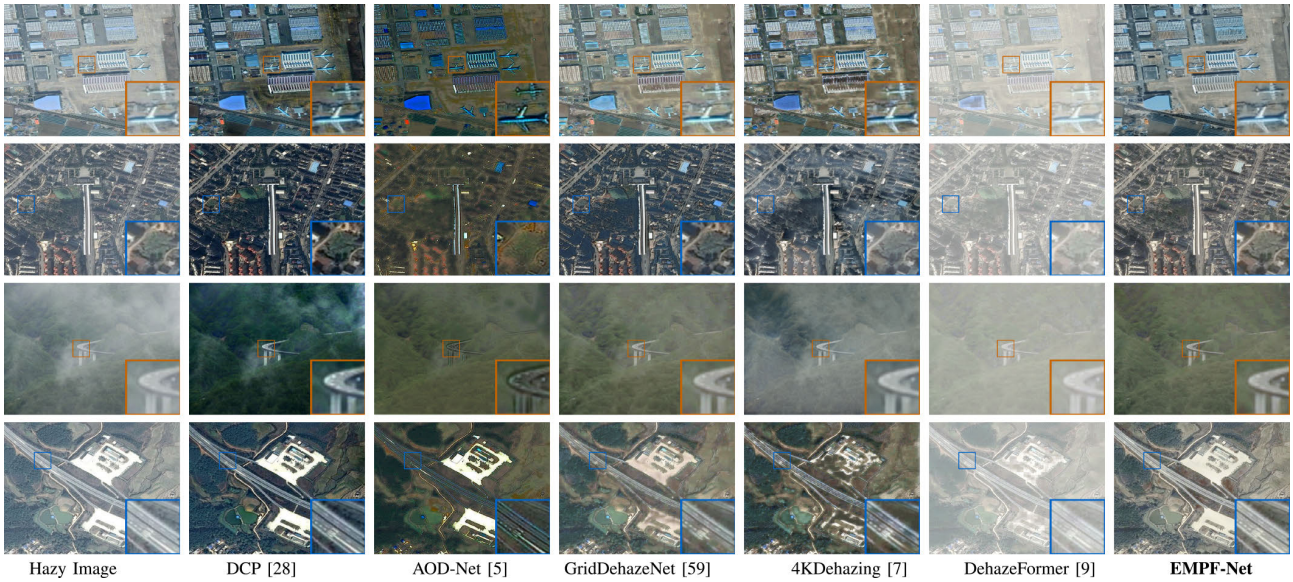



Fig. 9. Visual samples generated by the evaluated methods on RRSD300 datasets. Our EMPF-Net successfully eliminates haze and better restores image details, outperforming other evaluated methods.

positive manner. Each of these components significantly influences the final performance while maintaining computational efficiency, showcasing the effectiveness of our proposed architecture and design choices.

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TABLE IV

REAL-WORLD REMOTE SENSING IMAGE DEHAZING RESULTS. OUR PROPOSED EMPF-NET ACHIEVES THE BEST QUANTITATIVE PERFORMANCE ON RRSD300 DATASET

| Method | NIQE | IL-NIQE |
|---|---|---|
| DCP [28] | 4.590 | 15.37 |
| AOD-Net [5] | 5.515 | 15.82 |
| GridDehazeNet [59] | 5.117 | 15.74 |
| 4KDehazing [7] | 5.174 | 15.21 |
| DehazeFormer [9] | 8.351 | 16.77 |
| **EMPF-Net** | 4.247 | 14.11 |

TABLE V

ABLATION STUDY ON INDIVIDUAL COMPONENTS. EACH PROPOSED COMPONENT PLAYS A VALUABLE ROLE AND SHOWS A POSITIVE EFFECT ON THE OVERALL PERFORMANCE

| Model | PSNR | SSIM | MSE | #Param | FLOPs |
|---|---|---|---|---|---|
| Baseline | 23.20 | 0.8952 | 0.0047 | 1.202 M | 51.28 G |
| + EF | 23.51 | 0.8985 | 0.0045 | 1.202 M | 51.28 G |
| + MPLB | 22.41 | 0.8763 | 0.0061 | 289.1 K | 10.78 G |
| + MPQLB | 24.96 | 0.9099 | 0.0032 | 295.5 K | 11.07 G |
| + SCTAM | 25.89 | 0.9174 | 0.0026 | 303.1 K | 11.99 G |
| + PIFM | 26.33 | 0.9276 | 0.0023 | 300.2 K | 11.59 G |

TABLE VI

ABLATION STUDY ON INTRALEVEL FEATURE FUSION MODULES. OUR PROPOSED PIFM OBTAINS THE BEST PERFORMANCE OVER THE OPERATIONS OF CONVOLUTION AND SKFUSION

| Component | PSNR | SSIM | MSE | #Param | FLOPs |
|---|---|---|---|---|---|
| Convolution | 25.94 | 0.9174 | 0.0026 | 303.1 K | 11.99 G |
| SKFusion [9] | 26.12 | 0.9179 | 0.0025 | 298.5 K | 11.39 G |
| **PIFM** | 26.33 | 0.9276 | 0.0023 | 300.2 K | 11.59 G |

*2) Intralevel Feature Fusion:* To further evaluate the effectiveness of our proposed PIFM, we conduct comparative experiments on commonly used $1 \times 1$ convolution, SKFusion [9] and our PIFM. The results, as shown in Table VI, unequivocally demonstrate the superiority of our PIFM over traditional convolution, with a substantial improvement of 0.39 dB in PSNR. SKFusion performs slightly better than $1 \times 1$ convolution, which is consistent with the experimental findings reported in [9]. However, our PIFM surpasses SKFusion by an additional 0.21 dB in PSNR, highlighting the significant advantage of PIFM in remote sensing image dehazing. The performance comparisons clearly support our superior performance over traditional convolution and SKFusion, reinforcing the value of PIFM as a powerful physics-aware strategy for intralevel feature fusion.

*3) Basic Learning Blocks:* To assess the effectiveness of our proposed MPQLB, we conduct experiments and compare its performance with two commonly used learning blocks, namely RNB [63] and the FasterNetBlock (FNB) [17], in the context of remote sensing image dehazing. Table VII presents the results of our experiments, clearly demonstrating the superiority of our proposed MPQLB over the RNB and FNB. Notably, our MPQLB outperforms RNB by a margin of 1.32 dB and FNB by a margin of 0.66 dB in terms of PSNR. It is worth highlighting that our approach achieves these significant

TABLE VII

ABLATION STUDY ON BASIC LEARNING BLOCKS. OUR PROPOSED MPQLB DEMONSTRATES SUPERIOR PERFORMANCE WHILE SIGNIFICANTLY REDUCING THE NUMBER OF PARAMETERS AND COMPUTATIONAL COMPLEXITY

| Component | PSNR | SSIM | MSE | #Param | FLOPs |
|---|---|---|---|---|---|
| RNB [63] | 25.01 | 0.9084 | 0.0037 | 1.209 M | 52.02 G |
| FNB [17] | 25.67 | 0.9115 | 0.0029 | 485.0 K | 20.05 G |
| **MPQLB** | 26.33 | 0.9276 | 0.0023 | 300.2 K | 11.59 G |

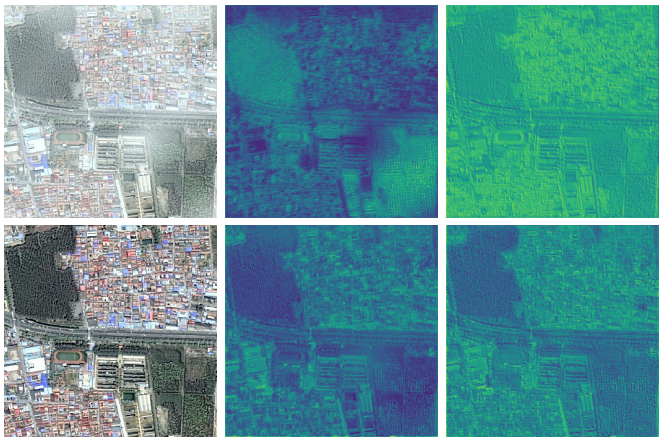


Fig. 10. Illustrations of the transmission maps and generated haze-free features in our proposed SCTAM. (Top) Hazy image, $1/\tilde{\mathbf{t}}$ in the first and second SCTAM. (Bottom) Ground truth, output features in the first and second SCTAM. The atmosphere light values corresponding to the first and second transmission maps are 0.0082 and 0.0237, respectively.

improvements while requiring the least #Param and FLOPs compared to the other two learning blocks. These findings unequivocally underscore the effectiveness of our MPQLB as a valuable improvement. The superior performance of MPQLB in terms of PSNR, along with its computational efficiency, solidifies its value in enhancing the overall performance of our proposed EMPF-Net for remote sensing image dehazing.

*4) What Does PIFM Learn?* Fig. 10 presents a visual representation of the transmission maps and output features within our proposed PIFM, where $1/\tilde{\mathbf{t}}$ represents the transmission map as defined in (7) and (10). The two groups of transmission maps have different resolutions, and we make adjustments to align them to improve the comparability. Following the atmospheric light scattering model [28], the bright regions in the transmission map indicate reduced light loss and lower levels of haze presence. Conversely, dark regions signify increased light loss and a higher density of haze in general. During the first PIFM in feature decoding network, distinct dark regions and haze reflections are visible within the transmission map. After applying (10), while a significant portion of the haze is effectively reduced, some remnants of haze persist in the output features. This empirical observation supports the claim that our PIFM demonstrates a strong grasp of physical principles. In contrast, in the last PIFM, nearly no haze reflections remain visible in the transmission map. This absence of dark regions indicates the gradual improvement of haze removal in feature decoding network, providing further evidence of the effectiveness of our PIFM. It also indicates

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TABLE VIII

ABLATION STUDY ON LOSS FUNCTIONS. THE HYBRID LOSS FUNCTION UTILIZED TO TRAIN OUR PROPOSED EMPF-NET OBTAINS THE BEST PERFORMANCE IN BOTH PSNR AND SSIM INDICATORS

| Loss Function | PSNR | SSIM | MSE |
|---|---|---|---|
| $\mathcal{L}_2$ | 26.14 | 0.9195 | 0.0025 |
| $\mathcal{L}_{perpetual}$ | 24.65 | 0.9300 | 0.0034 |
| $\mathcal{L}_2 + \lambda\mathcal{L}_{perpetual}$ | 26.33 | 0.9276 | 0.0023 |

the suppression of haze degradation as intralevel features from various sources are fused using our physics-aware PIFM. It is also worth noting that the atmospheric light value gradually increases from 0.0082 to 0.0237 corresponding to these two transmission maps, mirroring the luminance variation within the features. In summary, Fig. 10 serves as empirical validation of our PIFM's ability to acquire physics-aware features, enabling the decoupling and fusion of intralevel features from different sources. As a result, our PIFM effectively mitigates haze degradation while simultaneously facilitating intralevel feature fusion.

*5) Loss Functions:* Most dehazing methods [5], [7], [9] commonly employ the $\mathcal{L}_2$ loss function to train their networks. However, we have observed that using only the $\mathcal{L}_2$ loss function leads to a lower competitiveness of our model in terms of the SSIM indicator. We experimentally discover that perpetual loss $\mathcal{L}_{\text{perpetual}}$ shows a positive effect to the SSIM scores. As presented in Table VIII, while the $\mathcal{L}_2$ loss achieves the second-best results in terms of PSNR and mse, the SSIM score is lower than that obtained with the perpetual loss $\mathcal{L}_{\text{perpetual}}$. To strike a balance between the competing objectives, we employ a hybrid loss function $\mathcal{L}_2 + \lambda\mathcal{L}_{\text{perpetual}}$ to train our EMPF-Net. This hybrid approach effectively combines the strengths of both losses, resulting in the better overall performance.

### F. Efficiency Analysis

We conduct experiments to analyze the comparative efficiency of our proposed EMPF-Net against other evaluated learning-based methods. The results are presented in Table IX, where #Time represents the average processing duration for 100 haze-degraded images with a resolution of $3 \times 512 \times 512$ pixels. According to the findings, AOD-Net [5] exhibits the shortest inference time but obtains the lowest metrical scores, suggesting that it may compromise on performance to achieve faster processing. On the other hand, our proposed EMPF-Net ranks second in terms of processing time but achieves the best performance on remote sensing image dehazing. Notably, the processing time of our model is 43.92% faster compared to the current state-of-the-art DehazeFormer [9]. It is worth mentioning that our improved efficiency does not compromise on its ability to deliver superior dehazing results. These efficiency comparisons are visually represented in Fig. 1, which clearly highlights the faster and more effective nature of our approach. Our EMPF-Net stands out as a competitive solution, providing a remarkable balance between speed and performance, making it a practical and efficient choice for real-world application to eliminate haze degradation on remote sensing images.

TABLE IX

EFFICIENCY ANALYSIS RESULTS. OUR PROPOSED EMPF-NET ACHIEVES THE BEST TRADE-OFF BETWEEN THE QUALITY OF GENERATED HAZE-FREE IMAGES AND THE INFERENCE TIME TO PROCESS HAZY IMAGES

| Method | PSNR | SSIM | MSE | #Time | #Param | FLOPs |
|---|---|---|---|---|---|---|
| AOD-Net [5] | 15.92 | 0.7313 | 0.0265 | 0.0151 s | 1.761 K | 457.7 M |
| GridDehazeNet [59] | 20.24 | 0.8312 | 0.0096 | 0.0699 s | 955.7 K | 85.72 G |
| 4KDehazing [7] | 20.61 | 0.9264 | 0.0088 | 0.0549 s | 34.55 M | 105.8 G |
| DehazeFormer [9] | 24.64 | 0.9241 | 0.0034 | 0.0831 s | 1.205 M | 39.76 G |
| **EMPF-Net** | 26.33 | 0.9276 | 0.0023 | 0.0466 s | 300.2 K | 11.59 G |

## V. CONCLUSION

In this work, we introduce EMPF-Net, a light-weighted and efficient EMPF-Net for remote sensing image dehazing, which deviates from the traditional encoder–decoder framework, opting for an encoder-free architecture. The efficacy of EMPF-Net relies on the incorporation of three key components: MPQLB, SCTAM, and PIFM. These components play pivotal roles in augmenting feature learning and fusion processes. Despite having a remarkably low parameter count of 300k, our proposed model demonstrates significant advancements in both synthetic and real-world remote sensing dehazing scenarios. Our EMPF-Net achieves an optimal trade-off between inference time and image quality, thereby promoting dimension-wise queries, supervised cross-scale learning, and physics-aware fusion as new paradigms for remote sensing image dehazing. In future, our endeavors will focus on further optimizing the real-world performance and tackling the challenge of complex natural haze removal.

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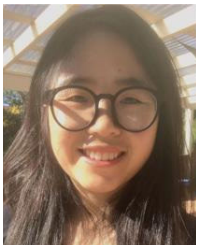
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